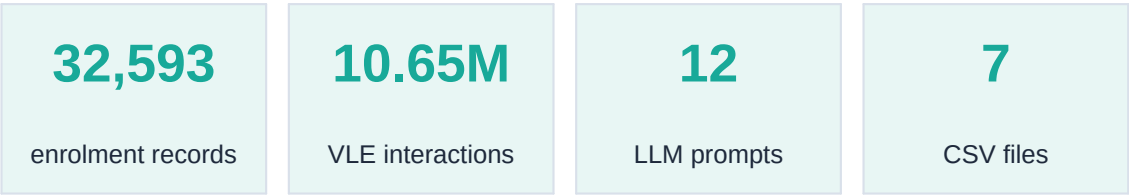


LLM Based Analysis of the OULAD Dataset

Prompt and output report for the LLM step

This document focuses only on the LLM analysis step. The prompts use verified summaries from the OULAD files, then the LLM provides interpretation, educational meaning, reliability controls, and limitations.



Scope of the LLM step

The LLM was not used as a raw calculator or database engine. It was given computed summaries, grouped values, missing value counts, feature descriptions, and selected statistics from the OULAD files. Its task was to explain what the evidence means and where interpretation must be cautious.

Method used in this report

Stage	Description
1	Prepare verified summaries from the OULAD CSV files.
2	Write targeted prompts around data quality, feature construction, engagement, demographics, and outcomes.
3	Record the prompt, supplied evidence, LLM output summary, and reliability control.
4	Keep numerical evidence separate from LLM interpretation.

1. LLM methodology and prompt index

The prompt design uses the same structure for every question: prompt used, evidence supplied, LLM output summary, and reliability control. This makes the LLM step transparent and limits unsupported interpretation.

Prompt	Analysis theme	LLM question
P1	Dataset coverage	What do the seven OULAD file shapes allow us to analyse?
P2	Target variable	What does final_result show, and when is passed useful?
P3	Missing socioeconomic data	How should missing imd_band values be interpreted?
P4	Missing assessment scores	How should missing score values be treated?
P5	Outliers	Are score and VLE click outliers errors or useful signals?
P6	Unit of analysis	Why use code_module, code_presentation, and id_student?
P7	Feature engineering	Which VLE and assessment features are useful for analysis?
P8	Master dataset	How should merged engineered values be interpreted?
P9	Categorical representation	How should categorical variables be represented?
P10	Demographic patterns	What do pass rates by gender and disability suggest?
P11	Engagement	How does VLE activity relate to final outcome?
P12	Correlation	Which engineered variables relate most to passing?

Evidence rule

Every number in this report comes from the uploaded OULAD CSV files or from derived summaries. LLM text is interpretation, not independent statistical evidence.

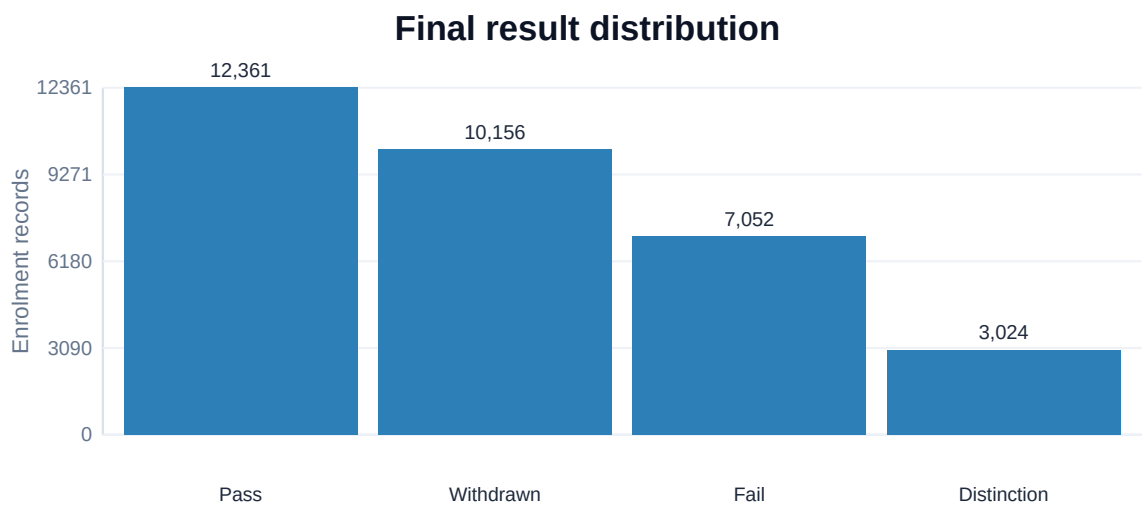
2. P1 to P2: dataset coverage and target variable

File	Rows	Columns
courses	22	3
assessments	206	6
studentAssessment	173,912	5
studentInfo	32,593	13
studentRegistration	32,593	5
vle	6,364	6
studentVle	10,655,280	6

P1 Data coverage and unit of analysis
Prompt used: Given the seven OULAD file shapes, what does dataset coverage allow us to analyse?
Evidence supplied: The dataset contains 32,593 studentInfo enrolment records, 28,785 unique students, 206 assessments, 6,364 VLE resources, and 10,655,280 VLE interaction rows.
LLM output summary: The dataset supports linked learning analytics across demographics, course enrolments, assessments, resources, and click activity. The enrolment record is safer than the student alone because a learner can appear in more than one module presentation.
Reliability control: File shapes confirm coverage but not quality. Joins, missing values, duplicates, and aggregation keys still need verification before interpretation.

2.2 P2: target variable interpretation

P2 Target variable and binary pass indicator
Prompt used: What does final_result show, and when is the binary passed feature useful or risky?
Evidence supplied: Final result counts are Pass 12,361, Withdrawn 10,156, Fail 7,052, and Distinction 3,024. The binary pass rate is 47.2%.
LLM output summary: The four outcome categories should remain visible because Withdrawn and Fail describe different learner situations. The binary passed field is useful for simple success analysis, but it hides the difference between academic failure and early withdrawal.
Reliability control: Binary simplification should not remove the original final_result column from interpretation.



3. P3 to P4: missing value interpretation

P3 Missing imd_band values
Prompt used: How should missing imd_band values be interpreted before analysis?
Evidence supplied: studentInfo has one missing column: imd_band has 1,111 missing values, equal to 3.41% of enrolment records.
LLM output summary: Missing socioeconomic information is also an equity issue. It should be preserved as Unknown or Not disclosed so that the missing group remains visible in later analysis.
Reliability control: Unknown is not the same as low deprivation. It should be interpreted as missing disclosure, not as a deprivation level.

P4 Missing assessment scores
Prompt used: How should missing assessment scores be handled and interpreted?
Evidence supplied: studentAssessment has 173 missing score values, equal to 0.10% of submissions. The median score in the score column is 80.
LLM output summary: Because the missing score share is very small, median filling can preserve the assessment table for feature creation. A missing score indicator can also be useful because absence of a score may reflect administrative issues, non submission, or late processing.
Reliability control: The LLM should not assume that a missing score means failure. The cause is not directly observed in the files.

Table	Field	Missing	LLM interpretation
studentInfo	imd_band	1,111	Keep as Unknown to preserve socioeconomic disclosure status.
studentAssessment	score	173	Small missing share, but missingness should still be documented.

4. P5: score and VLE click outliers

P5 | Outliers in assessment scores and VLE clicks

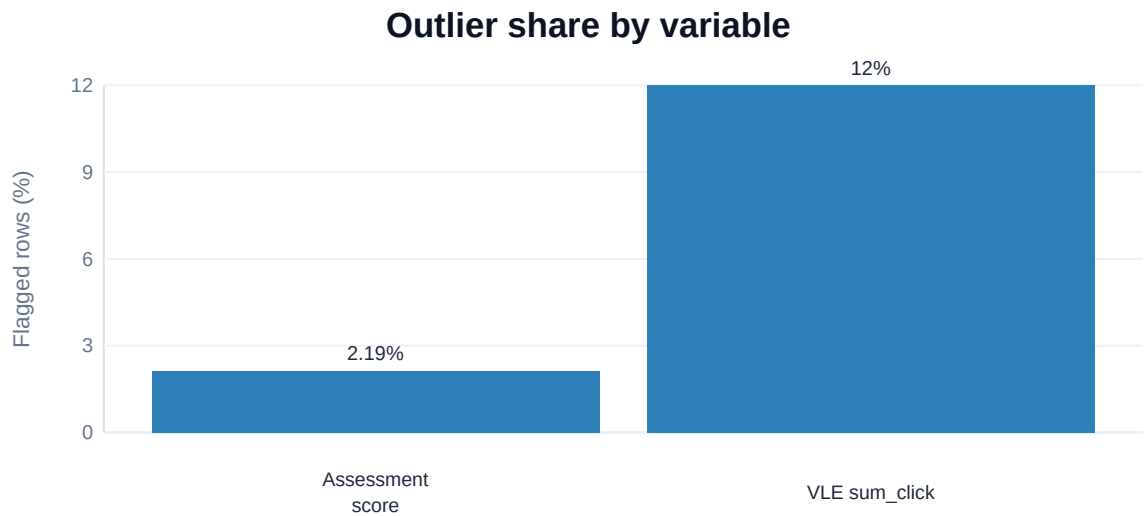
Prompt used: Are score and VLE click outliers data errors, or can they be useful signals?

Evidence supplied: Score IQR bounds are 27.5 to 127.5, with 3,813 flagged rows (2.19%). VLE sum_click bounds are minus 2.0 to 6.0, with 1,314,646 flagged rows (12.34%).

LLM output summary: Assessment score outliers may represent very weak performance or unusual records. VLE click outliers are especially meaningful because high click activity can indicate effort, confusion, repeated navigation, or resource intensive learning.

Reliability control: Outlier flags should support review and modelling diagnostics. They should not automatically remove learners from the dataset.

Variable	Q1	Q3	IQR	Flagged rows	Share
Assessment score	65.0	90.0	25.0	3,813	2.19%
VLE sum_click	1.0	3.0	2.0	1,314,646	12.34%



5. P6 to P7: enrolment key and engineered features

P6 Enrolment level key
Prompt used: Why should features be aggregated by code_module, code_presentation, and id_student?
Evidence supplied: studentInfo has 32,593 enrolment records and 28,785 unique students. The three column enrolment key also produces 32,593 unique records.
LLM output summary: The three column key keeps each learning experience separate. This prevents mixing the same learner across different module presentations and keeps each engineered feature aligned with one final result.
Reliability control: Grouping only by id_student can merge separate attempts or separate modules and distort outcome analysis.

P7 Core engineered features
Prompt used: Which VLE and assessment features are useful for analysing learning behaviour and performance?
Evidence supplied: VLE features are available for 29,228 enrolments and assessment features for 25,843 enrolments. Important features include total_clicks, avg_clicks_per_day, active_days, avg_score, num_assessments_submitted, and passed.
LLM output summary: These features convert raw logs and submissions into interpretable signals. VLE features describe engagement volume and rhythm. Assessment features describe academic participation and performance.
Reliability control: The features are proxies. High clicks do not prove high learning quality, and average score only exists for submitted assessments.

Feature family	Examples	LLM interpretation
VLE engagement	total_clicks, active_days, avg_clicks_per_day	Behavioural activity and course interaction.
Assessment behaviour	num_assessments_submitted , avg_score	Academic participation and performance.
Outcome	final_result, passed	Outcome interpretation and simplified success signal.

6. P8: master dataset construction

P8 Master dataset construction
Prompt used: How should merged engineered values be interpreted?
Evidence supplied: After assessment feature merging the shape is 32,593 by 15. After VLE feature merging the shape is 32,593 by 18. After encoding the shape is 32,593 by 48.
LLM output summary: Starting from all enrolment records keeps the full learner population. Filling missing engineered numeric values with zero is meaningful for features such as no observed submissions or no observed VLE activity, but the reason for absence must be interpreted carefully.
Reliability control: A zero in an engineered feature means no observed value in the available files. It should not automatically be interpreted as no effort or no motivation.

Dataset stage	Shape	Meaning
Base enrolment table	32,593 records	Full enrolment population.
After assessment features	32,593 by 15	Adds submission and score features.
After VLE features	32,593 by 18	Adds click based engagement features.
After categorical encoding	32,593 by 48	Creates model ready representation.

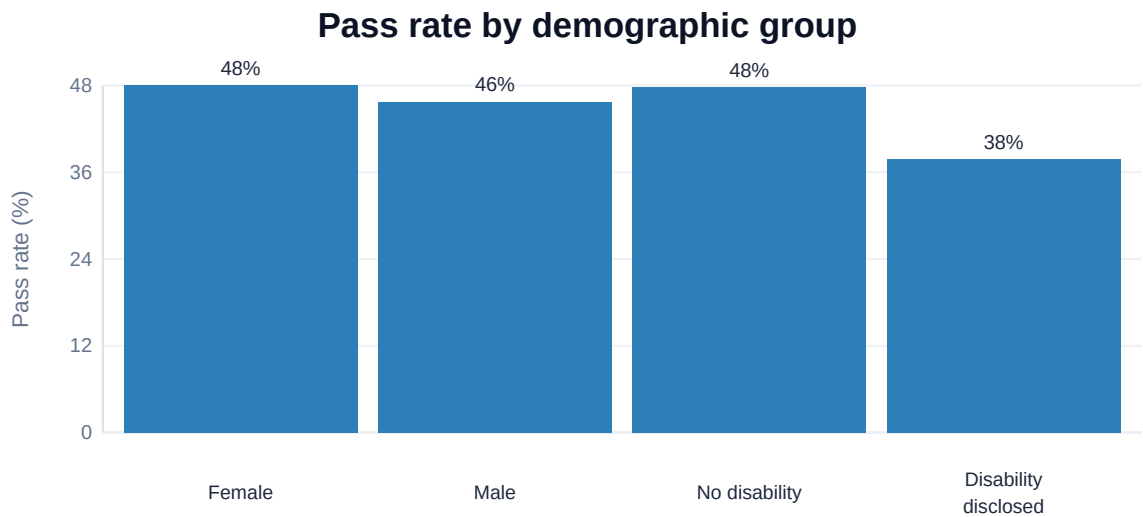
7. P9: categorical representation

P9 Categorical variable representation
Prompt used: How should categorical variables be represented for analysis and modelling?
Evidence supplied: Categorical cardinalities include code_module 7, code_presentation 4, gender 2, region 13, highest_education 5, imd_band 11 after Unknown handling, age_band 3, disability 2, and final_result 4.
LLM output summary: Binary variables can be encoded directly. Multi category variables such as region, age_band, highest_education, and imd_band require separate category representation to avoid creating artificial order.
Reliability control: Encoding improves model usability but may reduce readability. Any encoded features used in interpretation should be translated back to their original meaning.

Variable	Cardinality	LLM interpretation
gender	2	Simple binary representation is readable.
disability	2	Useful for accessibility analysis but sensitive in interpretation.
code_module	7	Course context should remain visible.
region	13	Separate category representation avoids false ranking.
imd_band	11	Unknown remains visible as disclosure status.
final_result	4	Keep the full outcome labels for interpretation.

8. P10: demographic pass rate patterns

P10 Demographic pass rate patterns
Prompt used: What do pass rates by gender and disability suggest after preprocessing?
Evidence supplied: Pass rates are female 48.44%, male 46.19%, no disability 48.18%, and disability disclosed 38.15%.
LLM output summary: The disability related gap is larger than the gender gap in this dataset. The LLM interprets this as a possible support and accessibility signal, not as a judgement about student ability.
Reliability control: These are descriptive associations. They do not prove that disability status causes the pass rate gap.



9. P11: VLE engagement by outcome

P11 | VLE engagement by outcome

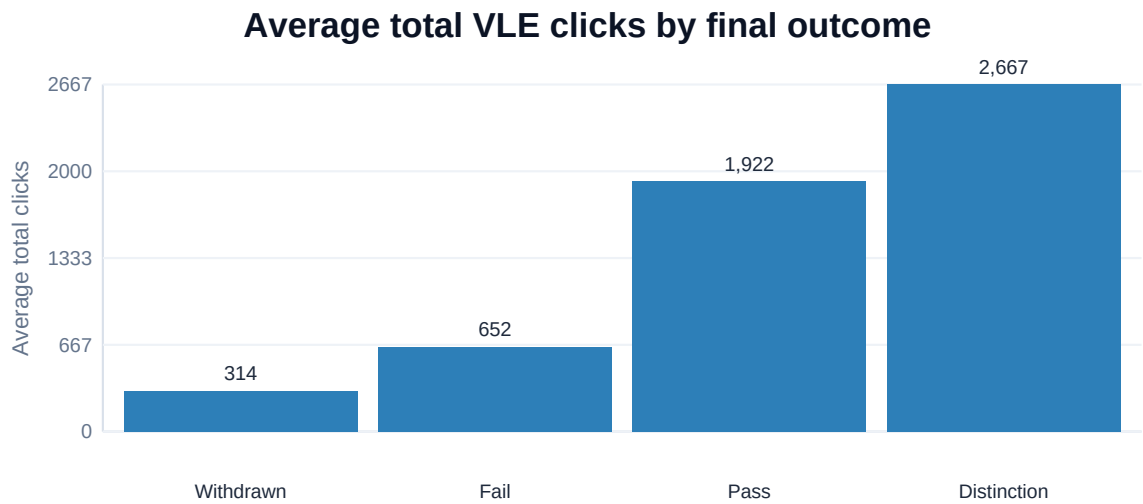
Prompt used: How does total VLE activity relate to final academic outcome?

Evidence supplied: Average total clicks are Withdrawn 313.95, Fail 651.85, Pass 1,921.81, and Distinction 2,666.76.

LLM output summary: Engagement increases clearly from Withdrawn to Distinction, so VLE activity is useful as a behavioural signal. However, clicks measure activity volume, not understanding or quality of learning.

Reliability control: Students can click often and still fail. Engagement should be interpreted with assessment score, submission count, timing, and course context.

Outcome	Average total clicks	LLM interpretation
Withdrawn	313.95	Very low observed engagement.
Fail	651.85	Higher than withdrawn but still low compared with successful outcomes.
Pass	1,921.81	Strong engagement signal.
Distinction	2,666.76	Highest engagement level.



10. P12: correlation analysis

P12 | Correlation analysis of engineered features

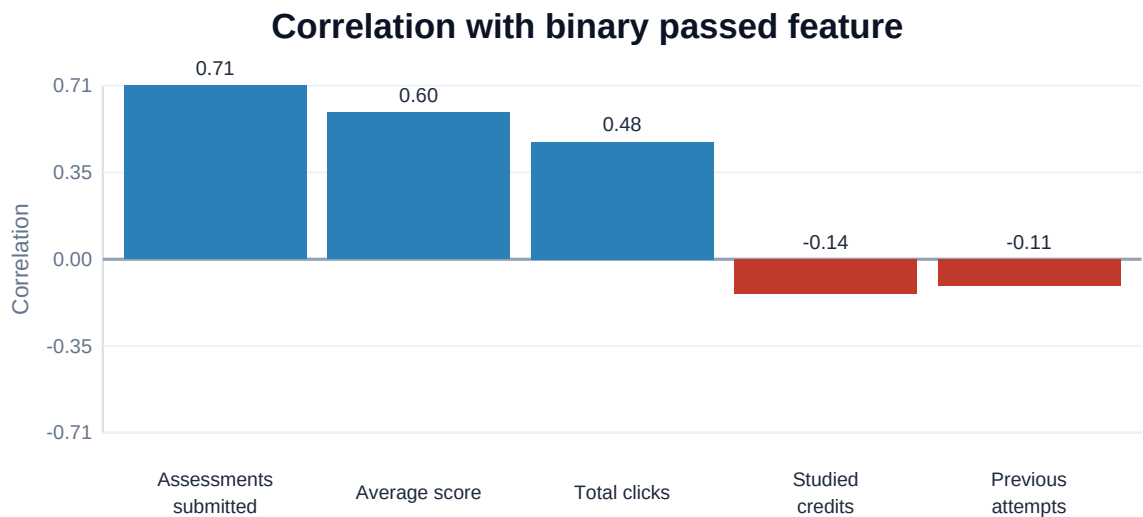
Prompt used: Which engineered variables are most related to the binary passed feature?

Evidence supplied: Correlations with passed are num_assessments_submitted 0.71, avg_score 0.60, total_clicks 0.48, studied_credits minus 0.14, and previous_attempts minus 0.11.

LLM output summary: Submission count and assessment score are the strongest positive indicators. VLE clicks are also relevant, while previous attempts and studied credits show weaker negative relationships in this summary.

Reliability control: Correlation is not causation. These values guide interpretation and feature selection, not causal intervention claims.

Feature	Correlation with passed	LLM interpretation
num_assessments_submitted	0.71	Strong participation signal.
avg_score	0.60	Direct performance signal.
total_clicks	0.48	Important engagement signal.
studied_credits	minus 0.14	Weak negative relationship.
previous_attempts	minus 0.11	Weak negative relationship.



11. Reliability, bias, and conclusion

The LLM step is useful when it is grounded in verified numerical evidence. Its main contribution is interpretation: it explains why missing values matter, why click activity should be treated carefully, how demographic patterns may create fairness concerns, and why correlations must not be read as causal proof.

Risk	Control used in this LLM report
Scalability	Large raw tables are summarized before prompting.
Hallucination	Every numerical claim is tied to supplied evidence.
Reproducibility	The prompt, evidence, output summary, and reliability control are documented.
Causal inference	Findings are described as associations unless causal testing is performed.
Demographic bias	Risk signals are framed as support needs, not as labels of student ability.
Interpretability	Evidence and interpretation are separated in each prompt card.

Conclusion

This LLM report shows how a language model can support educational data interpretation when it receives verified OULAD summaries. The strongest LLM contributions are fairness interpretation for missing socioeconomic data, careful reading of VLE engagement, cautious handling of outliers, and clear warnings against causal overclaiming.

The LLM step should therefore be used as an explanation and critical reflection layer. It helps transform computed evidence into educational meaning, while the numerical evidence itself must remain grounded in the dataset.